

Ninapro DB2 sEMG: EDA Summary

Dataset & Quality

- **Setup:** 20 subjects, 12 EMG channels sampled at 2000 Hz (Exercise 1)
- **Health:** Zero missing values and clean data; extreme values (7–18%) represent genuine muscle-contraction bursts rather than sensor noise, so they are kept
- **Challenge:** Heavy class imbalance (~70% "Rest" class) requiring balanced metrics like Macro F1 instead of raw accuracy.

Key Findings

- **Separability:** Visualizations (PCA, UMAP, boxplots) confirm distinct signal activation and amplitude differences across active gestures versus rest.
- **Cross-Talk:** A strong correlation (>0.90) among neighboring channels indicates electrode interference, a critical spatial characteristic to directly incorporate into the spatio-temporal GNN graph model.